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medgemma

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MedGemma is a collection of Gemma 3 variants that are trained for performance on medical text and image comprehension.

[4b](#) [27b](#)

Models

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Readme

MedGemma model card

Model documentation: [MedGemma](#)

Resources:

- Model on Google Cloud Model Garden: [MedGemma](#)
- Model on Hugging Face: [MedGemma](#)
- GitHub repository (supporting code, Colab notebooks, discussions, and issues): [MedGemma](#)
- Quick start notebook: [GitHub](#)

- Fine-tuning notebook: [GitHub](#)
- [Patient Education Demo built using MedGemma](#)
- Support: See [Contact](#)
- License: The use of MedGemma is governed by the [Health AI Developer Foundations terms of use](#).

Author: Google

Model information

This section describes the MedGemma model and how to use it.

Description

MedGemma is a collection of [Gemma 3](#) variants that are trained for performance on medical text and image comprehension. Developers can use MedGemma to accelerate building healthcare-based AI applications. MedGemma currently comes in two variants: a 4B multimodal version and a 27B text-only version.

MedGemma 4B utilizes a [SigLIP](#) image encoder that has been specifically pre-trained on a variety of de-identified medical data, including chest X-rays, dermatology images, ophthalmology images, and histopathology slides. Its LLM component is trained on a diverse set of medical data, including radiology images, histopathology patches, ophthalmology images, and dermatology images.

MedGemma 4B is available in both pre-trained (suffix: `-pt`) and instruction-tuned (suffix `-it`) versions. The instruction-tuned version is a better starting point for most applications. The pre-trained version is available for those who want to experiment more deeply with the models.

MedGemma 27B has been trained exclusively on medical text and optimized for inference-time computation. MedGemma 27B is only available as an instruction-tuned model.



details.

A full technical report will be available soon.

How to use

Below are some example code snippets to help you quickly get started running the model locally on GPU. If you want to use the model at scale, we recommend that you create a production version using [Model Garden](#).

First, install the Transformers library. Gemma 3 is supported starting from transformers 4.50.0.

```
$ pip install -U transformers
```

Run model with the pipeline API

```
from transformers import pipeline
from PIL import Image
import requests
import torch

pipe = pipeline(
    "image-text-to-text",
    model="google/medgemma-4b-it",
    torch_dtype=torch.bfloat16,
    device="cuda",
)

# Image attribution: Stillwaterising, CC0, via Wikimedia Commons
image_url = "https://upload.wikimedia.org/wikipedia/commons/c/c8/Chest_Xray_PA_3-8-2010.png"
image = Image.open(requests.get(image_url, headers={"User-Agent": "example"}, stream=True).raw)

messages = [
    {
        "role": "system",
        "content": [{"type": "text", "text": "You are an expert radiologist."}]
    },
    {
        "role": "user",
        "content": [
            {"type": "text", "text": "Describe this X-ray"},
            {"type": "image", "image": image},
        ]
    }
]

output = pipe(text=messages, max_new_tokens=200)
print(output[0]["generated_text"][-1]["content"])
```

Run the model directly

```
# pip install accelerate
from transformers import AutoProcessor, AutoModelForImageTextToText
from PIL import Image
import requests
import torch

model_id = "google/medgemma-4b-it"

model = AutoModelForImageTextToText.from_pretrained(
    model_id,
    torch_dtype=torch.bfloat16,
    device_map="auto",
)
processor = AutoProcessor.from_pretrained(model_id)

# Image attribution: Stillwaterising, CC0, via Wikimedia Commons
image_url = "https://upload.wikimedia.org/wikipedia/commons/c/c8/Chest_Xray_PA_3-8-2010.png"
image = Image.open(requests.get(image_url, headers={"User-Agent": "example"}, stream=True).raw)

messages = [
    {
        "role": "system",
        "content": [{"type": "text", "text": "You are an expert radiologist."}]
    },
    {
        "role": "user",
        "content": [
            {"type": "text", "text": "Describe this X-ray"},
            {"type": "image", "image": image}
        ]
    }
]

inputs = processor.apply_chat_template(
    messages, add_generation_prompt=True, tokenize=True,
    return_dict=True, return_tensors="pt"
).to(model.device, dtype=torch.bfloat16)

input_len = inputs["input_ids"].shape[-1]

with torch.inference_mode():
    generation = model.generate(*inputs, max_new_tokens=200, do_sample=False)
    generation = generation[0][input_len:]

decoded = processor.decode(generation, skip_special_tokens=True)
print(decoded)
```

Examples

See the following Colab notebooks for examples of how to use MedGemma:

- To give the model a quick try, running it locally with weights from Hugging Face, see [Quick start notebook in Colab](#). Note that you will need to use Colab Enterprise to run the 27B model without quantization.
- For an example of fine-tuning the model, see the [Fine-tuning notebook in Colab](#).

Model architecture overview

The MedGemma model is built based on [Gemma 3](#) and uses the same decoder-only transformer architecture as Gemma 3. To read more about the architecture, consult the Gemma 3 [model card](#).

Technical specifications

- **Model type:** Decoder-only Transformer architecture, see the [Gemma 3 technical report](#)
- **Modalities:** 4B: Text, vision; 27B: Text only
- **Attention mechanism:** Utilizes grouped-query attention (GQA)
- **Context length:** Supports long context, at least 128K tokens
- **Key publication:** Coming soon
- **Model created:** May 20, 2025
- **Model version:** 1.0.0

Citation

A technical report is coming soon. In the meantime, if you publish using this model, please cite the Hugging Face model page:

```
@misc{medgemma-hf,  
  author = {Google},  
  title = {MedGemma Hugging Face}  
  howpublished = {\url{https://huggingface.co/collections/google/medgemma-release-  
680aade845f90bec6a3f60c4}},  
  year = {2025},  
  note = {Accessed: [Insert Date Accessed, e.g., 2025-05-20]}}  
}
```

Inputs and outputs

Input:

- Text string, such as a question or prompt

- Images, normalized to 896 x 896 resolution and encoded to 256 tokens each
- Total input length of 128K tokens

Output:

- Generated text in response to the input, such as an answer to a question, analysis of image content, or a summary of a document
- Total output length of 8192 tokens

Performance and validation

MedGemma was evaluated across a range of different multimodal classification, report generation, visual question answering, and text-based tasks.

Key performance metrics

Imaging evaluations

The multimodal performance of MedGemma 4B was evaluated across a range of benchmarks, focusing on radiology, dermatology, histopathology, ophthalmology, and multimodal clinical reasoning.

MedGemma 4B outperforms the base Gemma 3 4B model across all tested multimodal health benchmarks.

Task and metric	MedGemma 4B	Gemma 3 4B
Medical image classification		
MIMIC CXR - Average F1 for top 5 conditions	88.9	81.1
CheXpert CXR - Average F1 for top 5 conditions	48.1	31.2
DermMCQA* - Accuracy	71.8	42.6
Visual question answering		
SlakeVQA (radiology) - Tokenized F1	62.3	38.6
VQA-Rad** (radiology) - Tokenized F1	49.9	38.6
PathMCQA (histopathology, internal***) - Accuracy	69.8	37.1
Knowledge and reasoning		
MedXpertQA (text + multimodal questions) - Accuracy	18.8	16.4

*Described in [Liu \(2020, Nature medicine\)](#), presented as a 4-way MCQ per example for skin condition classification.

****Based on "balanced split," described in [Yang \(2024, arXiv\)](#).**

*****Based on multiple datasets, presented as 3-9 way MCQ per example for identification, grading, and subtype for breast, cervical, and prostate cancer.**

Chest X-ray report generation

MedGemma chest X-ray (CXR) report generation performance was evaluated on [MIMIC-CXR](#) using the [RadGraph F1 metric](#). We compare the MedGemma pre-trained checkpoint with our previous best model for CXR report generation, [PaliGemma 2](#).

Metric	MedGemma 4B (pre-trained)	PaliGemma 2 3B (tuned for CXR)	PaliGemma 2 10B (tuned for CXR)
Chest X-ray report generation			
MIMIC CXR - RadGraph F1	29.5	28.8	29.5

The instruction-tuned versions of MedGemma 4B and Gemma 3 4B achieve lower scores (0.22 and 0.12, respectively) due to the differences in reporting style compared to the MIMIC ground truth reports. Further fine-tuning on MIMIC reports will enable users to achieve improved performance.

Text evaluations

MedGemma 4B and text-only MedGemma 27B were evaluated across a range of text-only benchmarks for medical knowledge and reasoning.

The MedGemma models outperform their respective base Gemma models across all tested text-only health benchmarks.

Metric	MedGemma 27B	Gemma 3 27B	MedGemma 4B	Gemma 3 4B
MedQA (4-op)	89.8 (best-of-5) 87.7 (0-shot)	74.9	64.4	50.7
MedMCQA	74.2	62.6	55.7	45.4
PubMedQA	76.8	73.4	73.4	68.4
MMLU Med (text only)	87.0	83.3	70.0	67.2
MedXpertQA (text only)	26.7	15.7	14.2	11.6
AfriMed-QA	84.0	72.0	52.0	48.0

For all MedGemma 27B results, [test-time scaling](#) is used to improve performance.

Ethics and safety evaluation

Evaluation approach

Our evaluation methods include structured evaluations and internal red-teaming testing of relevant content policies. Red-teaming was conducted by a number of different teams, each with different goals and human evaluation metrics. These models were evaluated against a number of different categories relevant to ethics and safety, including:

- **Child safety:** Evaluation of text-to-text and image-to-text prompts covering child safety policies, including child sexual abuse and exploitation.
- **Content safety:** Evaluation of text-to-text and image-to-text prompts covering safety policies, including harassment, violence and gore, and hate speech.
- **Representational harms:** Evaluation of text-to-text and image-to-text prompts covering safety policies, including bias, stereotyping, and harmful associations or inaccuracies.
- **General medical harms:** Evaluation of text-to-text and image-to-text prompts covering safety policies, including information quality and harmful associations or inaccuracies.

In addition to development level evaluations, we conduct “assurance evaluations” which are our “arms-length” internal evaluations for responsibility governance decision making. They are conducted separately from the model development team, to inform decision making about release. High-level findings are fed back to the model team, but prompt sets are held out to prevent overfitting and preserve the results’ ability to inform decision making. Notable assurance evaluation results are reported to our Responsibility & Safety Council as part of release review.

Evaluation results

For all areas of safety testing, we saw safe levels of performance across the categories of child safety, content safety, and representational harms. All testing was conducted without safety filters to evaluate the model capabilities and behaviors. For text-to-text, image-to-text, and audio-to-text, and across both MedGemma model sizes, the model produced minimal policy violations. A limitation of our evaluations was that they included primarily English language prompts.

Data card

Dataset overview

Training

The base Gemma models are pre-trained on a large corpus of text and code data. MedGemma 4B utilizes a [SigLIP](#) image encoder that has been specifically pre-trained on a variety of de-identified medical data, including radiology images, histopathology images, ophthalmology images, and dermatology images. Its LLM

component is trained on a diverse set of medical data, including medical text relevant to radiology images, chest-x rays, histopathology patches, ophthalmology images and dermatology images.

Evaluation

MedGemma models have been evaluated on a comprehensive set of clinically relevant benchmarks, including over 22 datasets across 5 different tasks and 6 medical image modalities. These include both open benchmark datasets and curated datasets, with a focus on expert human evaluations for tasks like CXR report generation and radiology VQA.

Source

MedGemma utilizes a combination of public and private datasets.

This model was trained on diverse public datasets including MIMIC-CXR (chest X-rays and reports), Slake-VQA (multimodal medical images and questions), PAD-UFES-20 (skin lesion images and data), SCIN (dermatology images), TCGA (cancer genomics data), CAMELYON (lymph node histopathology images), PMC-OA (biomedical literature with images), and Mendeley Digital Knee X-Ray (knee X-rays).

Additionally, multiple diverse proprietary datasets were licensed and incorporated (described next).

Data Ownership and Documentation

- **Mimic-CXR**: MIT Laboratory for Computational Physiology and Beth Israel Deaconess Medical Center (BIDMC).
- **Slake-VQA**: The Hong Kong Polytechnic University (PolyU), with collaborators including West China Hospital of Sichuan University and Sichuan Academy of Medical Sciences / Sichuan Provincial People's Hospital.
- **PAD-UFES-20**: Federal University of Espírito Santo (UFES), Brazil, through its Dermatological and Surgical Assistance Program (PAD).
- **SCIN**: A collaboration between Google Health and Stanford Medicine.
- **TCGA** (The Cancer Genome Atlas): A joint effort of National Cancer Institute and National Human Genome Research Institute. Data from TCGA are available via the Genomic Data Commons (GDC)
- **CAMELYON**: The data was collected from Radboud University Medical Center and University Medical Center Utrecht in the Netherlands.
- **PMC-OA (PubMed Central Open Access Subset)**: Maintained by the National Library of Medicine (NLM) and National Center for Biotechnology Information (NCBI), which are part of the NIH.
- **MedQA**: This dataset was created by a team of researchers led by Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits
- **Mendeley Digital Knee X-Ray**: This dataset is from Rani Channamma University, and is hosted on Mendeley Data.

- **AfriMed-QA**: This data was developed and led by multiple collaborating organizations and researchers include key contributors: Intron Health, SisonkeBiotik, BioRAMP, Georgia Institute of Technology, and MasakhaneNLP.
- **VQA-RAD**: This dataset was created by a research team led by Jason J. Lau, Soumya Gayen, Asma Ben Abacha, and Dina Demner-Fushman and their affiliated institutions (the US National Library of Medicine and National Institutes of Health)
- **MedExpQA**: This dataset was created by researchers at the HiTZ Center (Basque Center for Language Technology and Artificial Intelligence).
- **MedXpertQA**: This dataset was developed by researchers at Tsinghua University (Beijing, China) and Shanghai Artificial Intelligence Laboratory (Shanghai, China).

In addition to the public datasets listed above, MedGemma was also trained on de-identified datasets licensed for research or collected internally at Google from consented participants.

- Radiology dataset 1: De-identified dataset of different CT studies across body parts from a US-based radiology outpatient diagnostic center network.
- Ophthalmology dataset 1: De-identified dataset of fundus images from diabetic retinopathy screening.
- Dermatology dataset 1: De-identified dataset of teledermatology skin condition images (both clinical and dermatoscopic) from Colombia.
- Dermatology dataset 2: De-identified dataset of skin cancer images (both clinical and dermatoscopic) from Australia.
- Dermatology dataset 3: De-identified dataset of non-diseased skin images from an internal data collection effort.
- Pathology dataset 1: De-identified dataset of histopathology H&E whole slide images created in collaboration with an academic research hospital and biobank in Europe. Comprises de-identified colon, prostate, and lymph nodes.
- Pathology dataset 2: De-identified dataset of lung histopathology H&E and IHC whole slide images created by a commercial biobank in the United States.
- Pathology dataset 3: De-identified dataset of prostate and lymph node H&E and IHC histopathology whole slide images created by a contract research organization in the United States.
- Pathology dataset 4: De-identified dataset of histopathology, predominantly H&E whole slide images created in collaboration with a large, tertiary teaching hospital in the United States. Comprises a diverse set of tissue and stain types, predominantly H&E.

Data citation

- **MIMIC-CXR** Johnson, A., Pollard, T., Mark, R., Berkowitz, S., & Horng, S. (2024). MIMIC-CXR Database (version 2.1.0). PhysioNet. <https://physionet.org/content/mimic-cxr/2.1.0/> and Johnson, Alistair E. W., Tom J. Pollard, Seth J. Berkowitz, Nathaniel R. Greenbaum, Matthew P. Lungren, Chih-Ying Deng, Roger G.

- Mark, and Steven Horng. 2019. "MIMIC-CXR, a de-Identified Publicly Available Database of Chest Radiographs with Free-Text Reports." *Scientific Data* 6 (1): 1–8.
- **SLAKE** Liu, Bo, Li-Ming Zhan, Li Xu, Lin Ma, Yan Yang, and Xiao-Ming Wu. 2021. SLAKE: A Semantically-Labeled Knowledge-Enhanced Dataset for Medical Visual Question Answering." <http://arxiv.org/abs/2102.09542>.
 - **PAD-UEFS** Pacheco, A. G. C., Lima, G. R., Salomao, A., Krohling, B., Biral, I. P., de Angelo, G. G., Alves, F. O. G., Ju X. M., & P. R. C. (2020). PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. In *Proceedings of the 2020 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 1551-1558). IEEE. <https://doi.org/10.1109/BIBM49941.2020.9313241>
 - **SCIN** Ward, Abbi, Jimmy Li, Julie Wang, Sriram Lakshminarasimhan, Ashley Carrick, Bilson Campana, Jay Hartford, et al. 2024. "Creating an Empirical Dermatology Dataset Through Crowdsourcing With Web Search Advertisements." *JAMA Network Open* 7 (11): e2446615–e2446615.
 - **TCGA** The results shown here are in whole or part based upon data generated by the TCGA Research Network: <https://www.cancer.gov/tcga>.
 - **CAMELYON16** Ehteshami Bejnordi, Babak, Mitko Veta, Paul Johannes van Diest, Bram van Ginneken, Nico Karssemeijer, Geert Litjens, Jeroen A. W. M. van der Laak, et al. 2017. "Diagnostic Assessment of Deep Learning Algorithms for Detection of Lymph Node Metastases in Women With Breast Cancer." *JAMA* 318 (22): 2199–2210.
 - **MedQA** Jin, Di, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. 2020. "What Disease Does This Patient Have? A Large-Scale Open Domain Question Answering Dataset from Medical Exams." <http://arxiv.org/abs/2009.13081>.
 - **Mendeley Digital Knee X-Ray** Gornale, Shivanand; Patravali, Pooja (2020), "Digital Knee X-ray Images", Mendeley Data, V1, doi: 10.17632/t9ndx37v5h.1
 - **AfrimedQA** Olatunji, Tobi, Charles Nimo, Abraham Owodunni, Tassallah Abdullahi, Emmanuel Ayodele, Mardhiyah Sanni, Chinemelu Aka, et al. 2024. "AfriMed-QA: A Pan-African, Multi-Specialty, Medical Question-Answering Benchmark Dataset." <http://arxiv.org/abs/2411.15640>.
 - **VQA-RAD** Lau, Jason J., Soumya Gayen, Asma Ben Abacha, and Dina Demner-Fushman. 2018. "A Dataset of Clinically Generated Visual Questions and Answers about Radiology Images." *Scientific Data* 5 (1): 1–10.
 - **MedexpQA** Alonso, I., Oronoz, M., & Agerri, R. (2024). MedExpQA: Multilingual Benchmarking of Large Language Models for Medical Question Answering. *arXiv preprint arXiv:2404.05590*. Retrieved from <https://arxiv.org/abs/2404.05590>
 - **MedXpertQA** Zuo, Yuxin, Shang Qu, Yifei Li, Zhangren Chen, Xuekai Zhu, Ermo Hua, Kaiyan Zhang, Ning Ding, and Bowen Zhou. 2025. "MedXpertQA: Benchmarking Expert-Level Medical Reasoning and

Understanding." <http://arxiv.org/abs/2501.18362>.

De-identification/anonymization:

Google and partnerships utilize datasets that have been rigorously anonymized or de-identified to ensure the protection of individual research participants and patient privacy

Implementation information

Details about the model internals.

Software

Training was done using [JAX](#).

JAX allows researchers to take advantage of the latest generation of hardware, including TPUs, for faster and more efficient training of large models.

Use and limitations

Intended use

MedGemma is an open multimodal generative AI model intended to be used as a starting point that enables more efficient development of downstream healthcare applications involving medical text and images.

MedGemma is intended for developers in the life sciences and healthcare space. Developers are responsible for training, adapting and making meaningful changes to MedGemma to accomplish their specific intended use. MedGemma models can be fine-tuned by developers using their own proprietary data for their specific tasks or solutions.

MedGemma is based on Gemma 3 and has been further trained on medical images and text. MedGemma enables further development in any medical context (image and textual), however the model was pre-trained using chest X-ray, pathology, dermatology, and fundus images. Examples of tasks within MedGemma's training include visual question answering pertaining to medical images, such as radiographs, or providing answers to textual medical questions. Full details of all the tasks MedGemma has been evaluated can be found in an upcoming technical report.

Benefits

- Provides strong baseline medical image and text comprehension for models of its size.
- This strong performance makes it efficient to adapt for downstream healthcare-based use cases, compared to models of similar size without medical data pre-training.
- This adaptation may involve prompt engineering, grounding, agentic orchestration or fine-tuning depending on the use case, baseline validation requirements, and desired performance characteristics.

Limitations

MedGemma is not intended to be used without appropriate validation, adaptation and/or making meaningful modification by developers for their specific use case. The outputs generated by MedGemma are not intended to directly inform clinical diagnosis, patient management decisions, treatment recommendations, or any other direct clinical practice applications. Performance benchmarks highlight baseline capabilities on relevant benchmarks, but even for image and text domains that constitute a substantial portion of training data, inaccurate model output is possible. All outputs from MedGemma should be considered preliminary and require independent verification, clinical correlation, and further investigation through established research and development methodologies.

MedGemma's multimodal capabilities have been primarily evaluated on single-image tasks. MedGemma has not been evaluated in use cases that involve comprehension of multiple images.

MedGemma has not been evaluated or optimized for multi-turn applications.

MedGemma's training may make it more sensitive to the specific prompt used than Gemma 3.

When adapting MedGemma developer should consider the following:

- **Bias in validation data:** As with any research, developers should ensure that any downstream application

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inadvertently seen related medical information during its pre-training, potentially overestimating its true ability to generalize to novel medical concepts. Developers should validate MedGemma on datasets not publicly available or otherwise made available to non-institutional researchers to mitigate this risk.